

Eastern Mediterranean University – Computer Engineering Department

CMPE/CMSE/AING231 – Data Structures

Assignment #2

Prepared by: Assoc. Prof. Dr. Yiltan Bitirim

As a computer user, Alice wants to enter a set of 25 unique numbers as input and then search for a specific number in this set. To do this, write a C program for Alice by considering the following points.

1. You must keep the numbers entered as inputs in an array.
2. You should definitely use the **binary search** method to search the number (given by the user) in the array.
3. You must search with a recursive function that searches with the binary search method.
4. Whether you find the searched number in the series or not, you must inform the user with an appropriate message.
5. The binary search method works in arrays whose elements are ORDERED. The set of numbers that the user enters as input may not be ordered. After placing the elements in the array, check whether the array elements are in order. If it is in order, proceed to search. If it is not ordered, sort the array elements **using the Merge Sort** or **Quick Sort** and then proceed to the search. You must implement the sort algorithm with recursion.

NOTE:

If any of the following situations occur, you will get ZERO:

- You have not uploaded your solution by the deadline.
- You have produced a solution other than C language.
- You have uploaded an incorrect file or a file that cannot be opening.
- You have not used recursive functions as mentioned above.
- Your solution contains even a single line of code that is NOT yours.